

Replacing Enterprise Software with a Purpose-Built Operations Platform

Healthcare-Adjacent Operations | Full Stack | AWS

Overview

A healthcare-adjacent organization was managing daily operations across two enterprise platforms, multiple spreadsheets, manual document workflows, and disconnected accounting processes. The fragmentation created administrative overhead, duplicated data entry, compliance exposure, and limited visibility into case status across the organization.

They needed to replace this infrastructure with a single purpose-built platform designed around how the business actually operates. I was brought in as the sole engineer to design, build, and deliver it.

The Challenge

The project required replacing functionality previously spread across two enterprise platforms, consolidating deeply manual workflows into a unified system, and building a secure dual-portal architecture supporting both internal operational staff and a large external user network — all within a HIPAA-conscious infrastructure, with limited technical documentation from existing systems, and sole engineering responsibility across every layer of the stack.

Additional constraints included:

- HIPAA-aligned infrastructure and access controls
- Secure document handling for sensitive case records
- Complex relational workflows spanning cases, contacts, documents, services, and financial transactions
- External portal access for a large network of outside users
- Migration away from heavily manual processes
- Full ownership across frontend, backend, infrastructure, and all integrations

The Solution

I designed and built a custom web-based platform consolidating all core business workflows into a single application, supporting both internal staff and external users through distinct authenticated interfaces.

The platform included role-based access control, case and contact management, secure document storage and retrieval, workflow and status tracking, financial reporting, administrative tooling, external accounting integration, and audit-oriented operational logging.

Infrastructure was deployed on AWS with encrypted storage and transport, VPC network isolation, KMS key management, PHI access logging, restricted network access, controlled document access patterns, and centralized environment configuration, built to support the organization's compliance requirements as it scales.

Tech Stack

React, Next.js, Node.js, PostgreSQL, AWS (EC2, RDS, S3, SES, IAM, KMS, CloudWatch), Auth0, Tailwind CSS

Outcome

Over an 18-month engagement I took the platform from initial architecture through production deployment and internal testing, owning every layer of the stack throughout.

- Built a 70+ table relational data model consolidating core operational workflows
- Designed dual portal architecture serving distinct internal and external user populations
- Fully replaced the organization's primary operations platform, including a custom migration pipeline across 6 entity types
- Migration included deduplication, merge logic, field resolution, association mapping, and document transfer
- Final import involved ingesting hundreds of thousands of records and tens of thousands of files into production
- Integrated an external accounting API to handle a complex multi-account payment process
- Reduced manual administrative handling across document, case, and financial workflows
- Deployed HIPAA-conscious cloud infrastructure to production